

Laser Beam Melting of Lunar Regolith Simulant Under Lunar Conditions

System Progress and Initial Results

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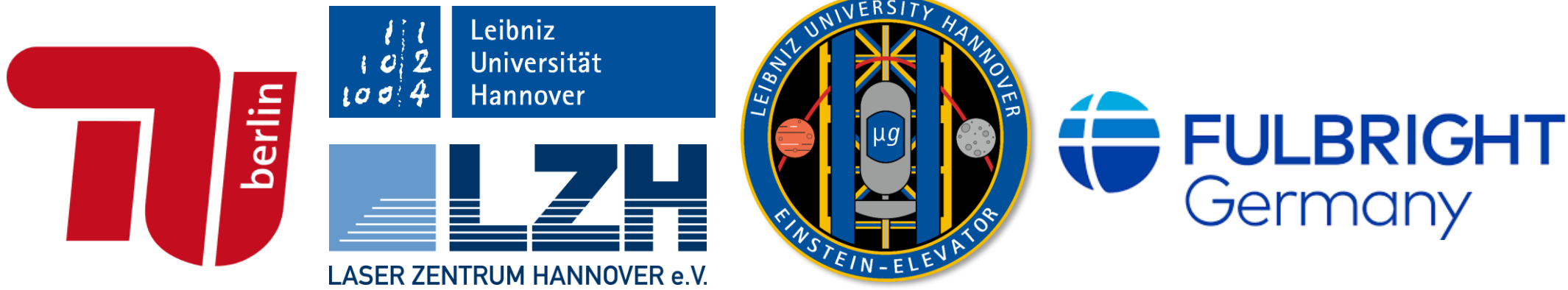
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Question

How does the reduced gravity and the lack of atmosphere on the surface of the Moon affect laser beam melted lunar regolith production processes and products?

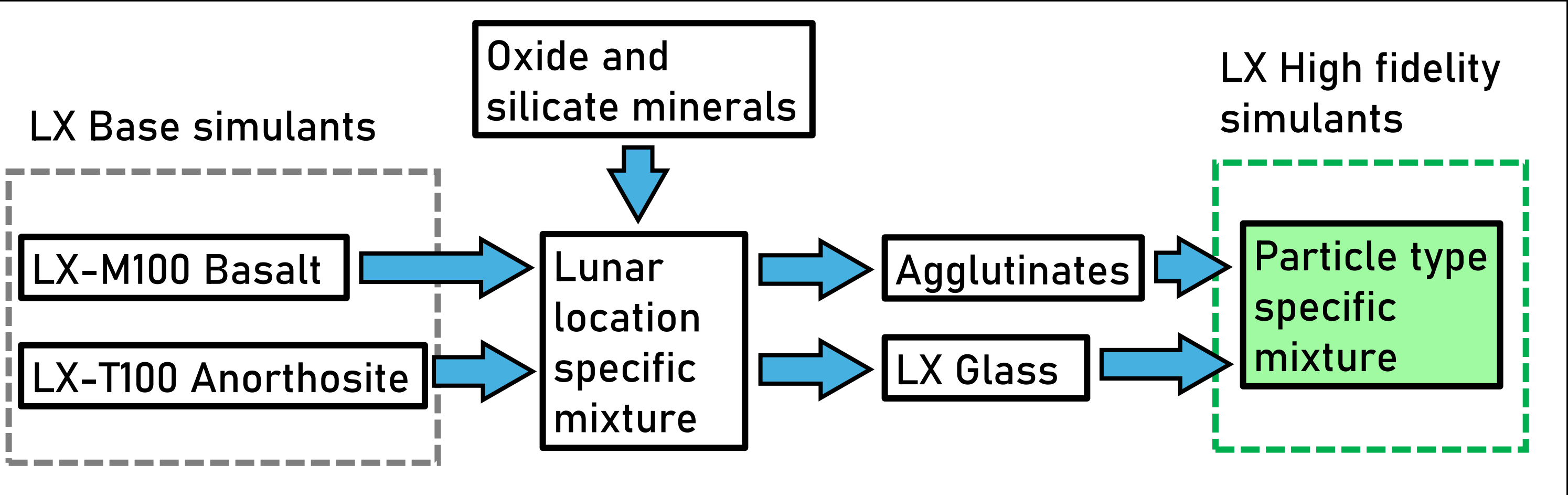


Simulated in Experiments on Earth

Lunar regolith: LX lunar regolith simulant system
Lunar gravity (1.62 m/s^2): Einstein-Elevator drop tower
Lunar vacuum ($3 \cdot 10^{-15} \text{ bar}$): Vacuum chamber

Regolith Simulant

- LX lunar regolith simulant system, developed at TU Berlin
- Designed to match lunar regolith chemistry, mineralogy, and physical properties
- Base simulants: Mare (basalt) and Terrae (anorthosite)
- High fidelity additives: olivine, ilmenite, glasses, and agglutinates [3]

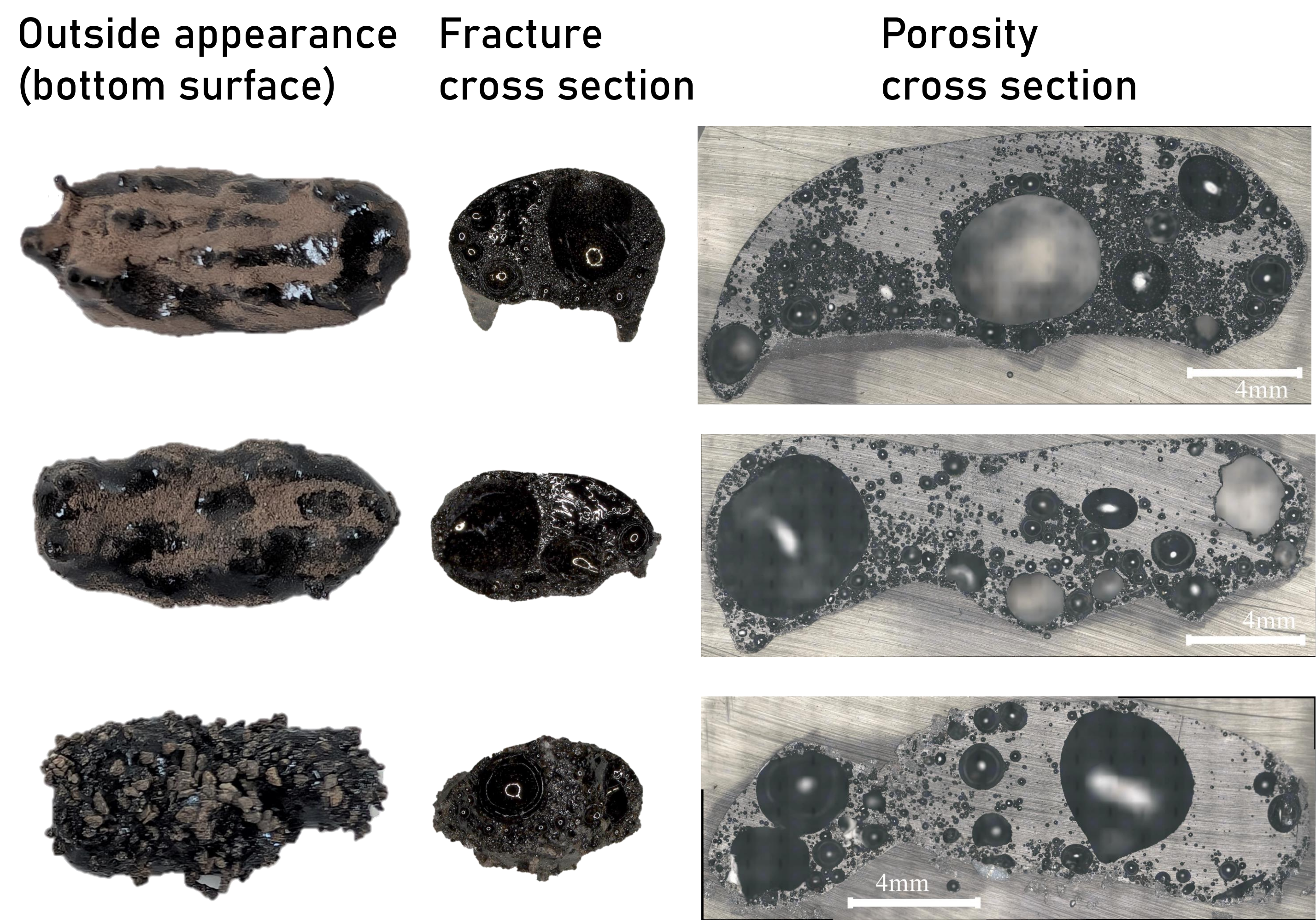


▲ LX lunar regolith simulant system allows for production and use of simulants of varying compositions, increasing in fidelity from left to right [3]

Current Areas of Investigation

- Vacuum: Varying ambient pressure across laser beam melting trials of LX-I50 regolith simulant from 0.05 to 2,000 mbar found that lower pressures led to samples with higher porosities (by factor of 3) and lower strengths (by factor of 2) [4]
- Particle size: Initial results from samples fabricated from varying particle sizes (<45, 90-160, and 500-1,000 μm) of LX-M regolith simulant suggest that larger particle sizes lead to lower porosities
- Gravity: Single-layer samples fabricated from various compositions of LX base and high-fidelity simulants under reduced gravity conditions are being analyzed, and will be compared to samples fabricated under earth gravity

Initial Results



▲ Samples fabricated from LX-M regolith simulant using varying particle sizes: <45 μm (top), 90-160 μm (center), and 500-1,000 μm (bottom); with laser parameters: power 200 W, scanning speed 2 mm/s, hatch distance 1mm, and spot size 1.8 mm

Funding

This project is funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under project number 458595944. The research by H. Fisher at TU Berlin is funded by the Deutsch-Amerikanische Fulbright-Kommission (German-American Fulbright Commission), through the 2024-2025 Open Study/Research Award.

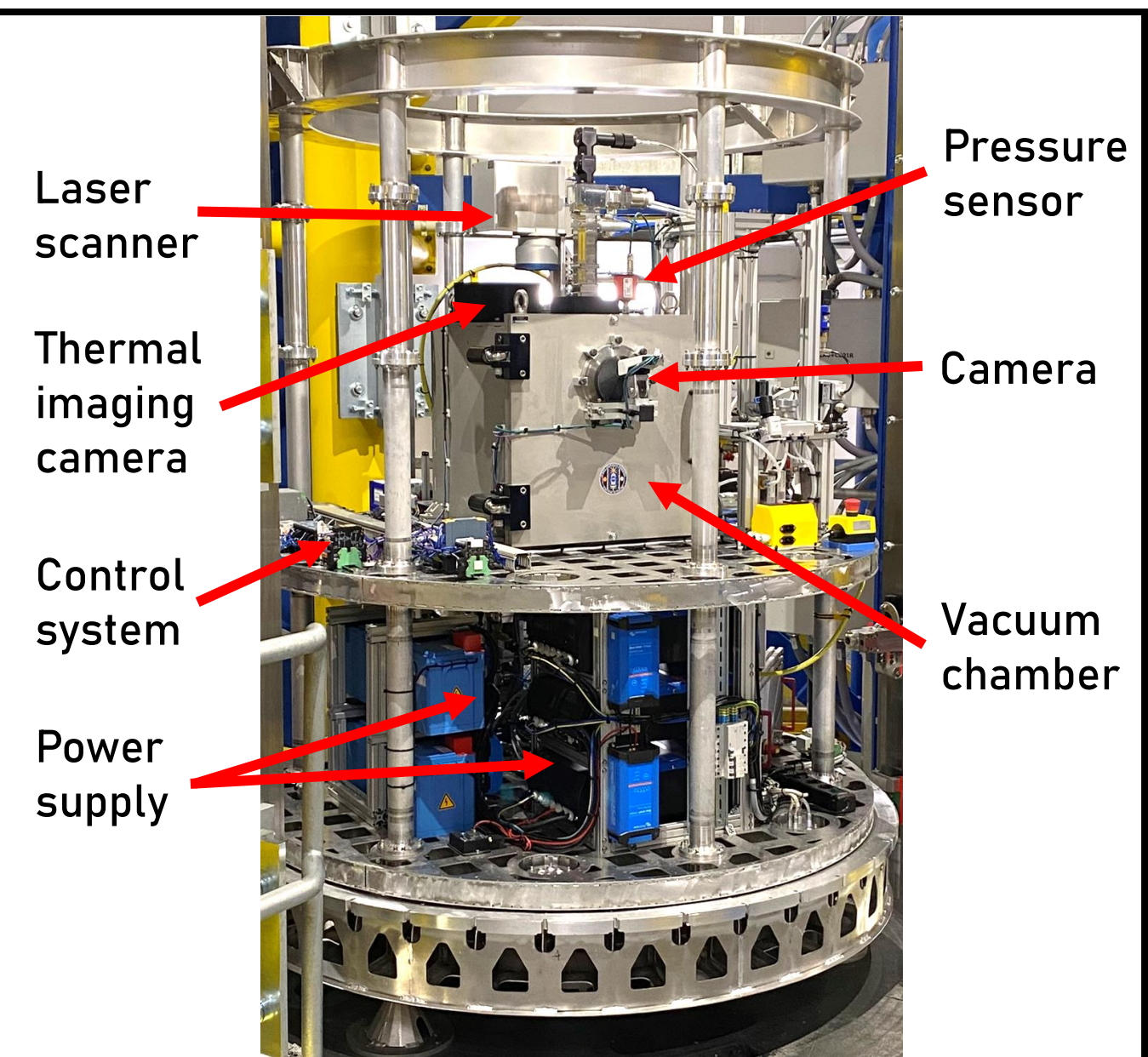
Einstein-Elevator and Experimental Setup

Overview

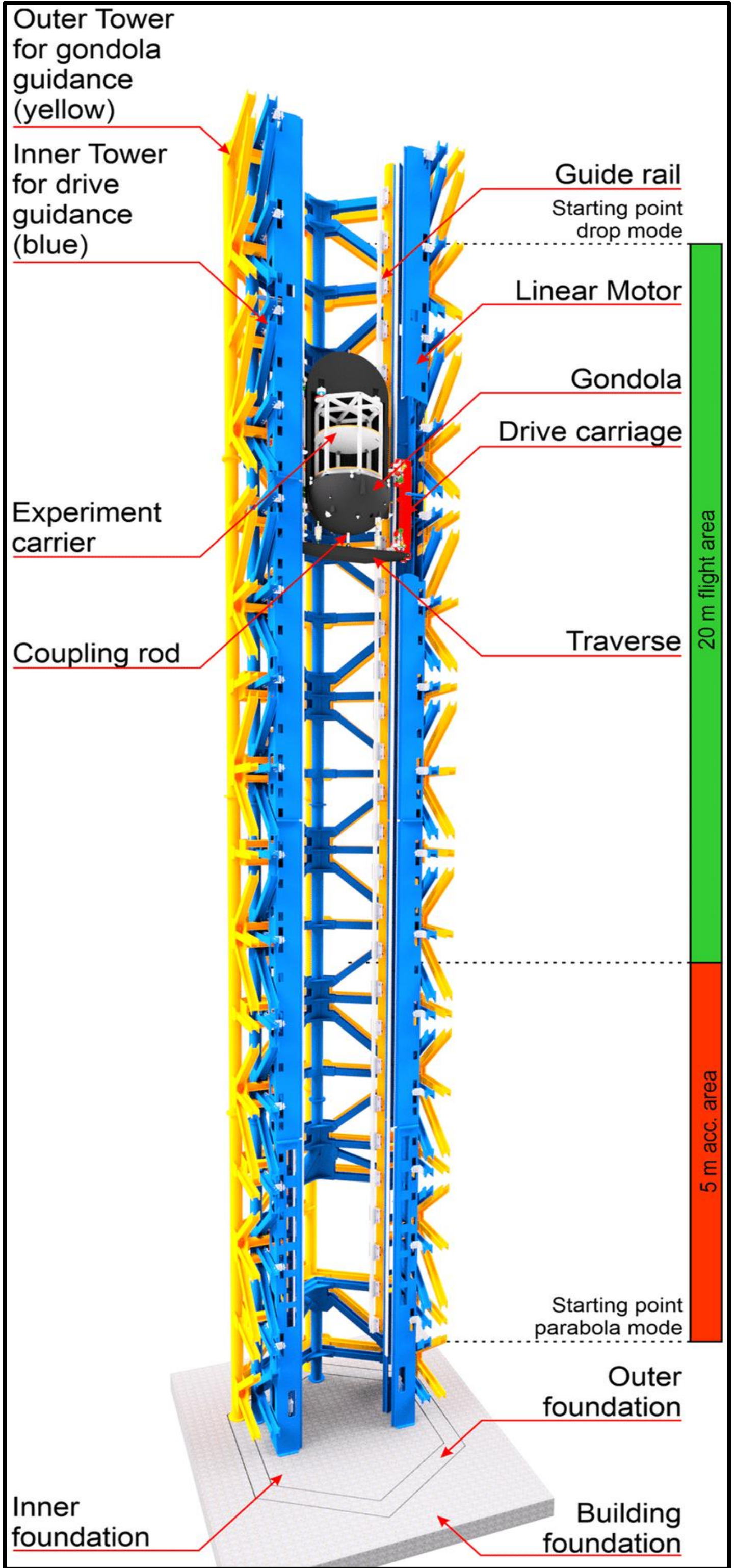
- Drop tower that enables experimentation at conditions ranging from microgravity to $5g$
- Experiments within a vacuum chamber on the elevator gondola allow for simultaneous simulation of lunar gravity and atmosphere
- Located in Hannover, Germany
- Operational since 2019

Technical Specifications

- Payload capacity of 1,000 kg, $\phi 1.7 \text{ m} \times 2.0 \text{ m}$
- Free fall distance of 20 m
- Vacuum strength of 10^{-2} mbar
- Up to 300 experiments per day
- Accelerations from $0g - 5g$, to simulate gravity on the Moon, Mars, and during rocket launches
- 4 seconds of lunar gravity per experiment run [1, 2]



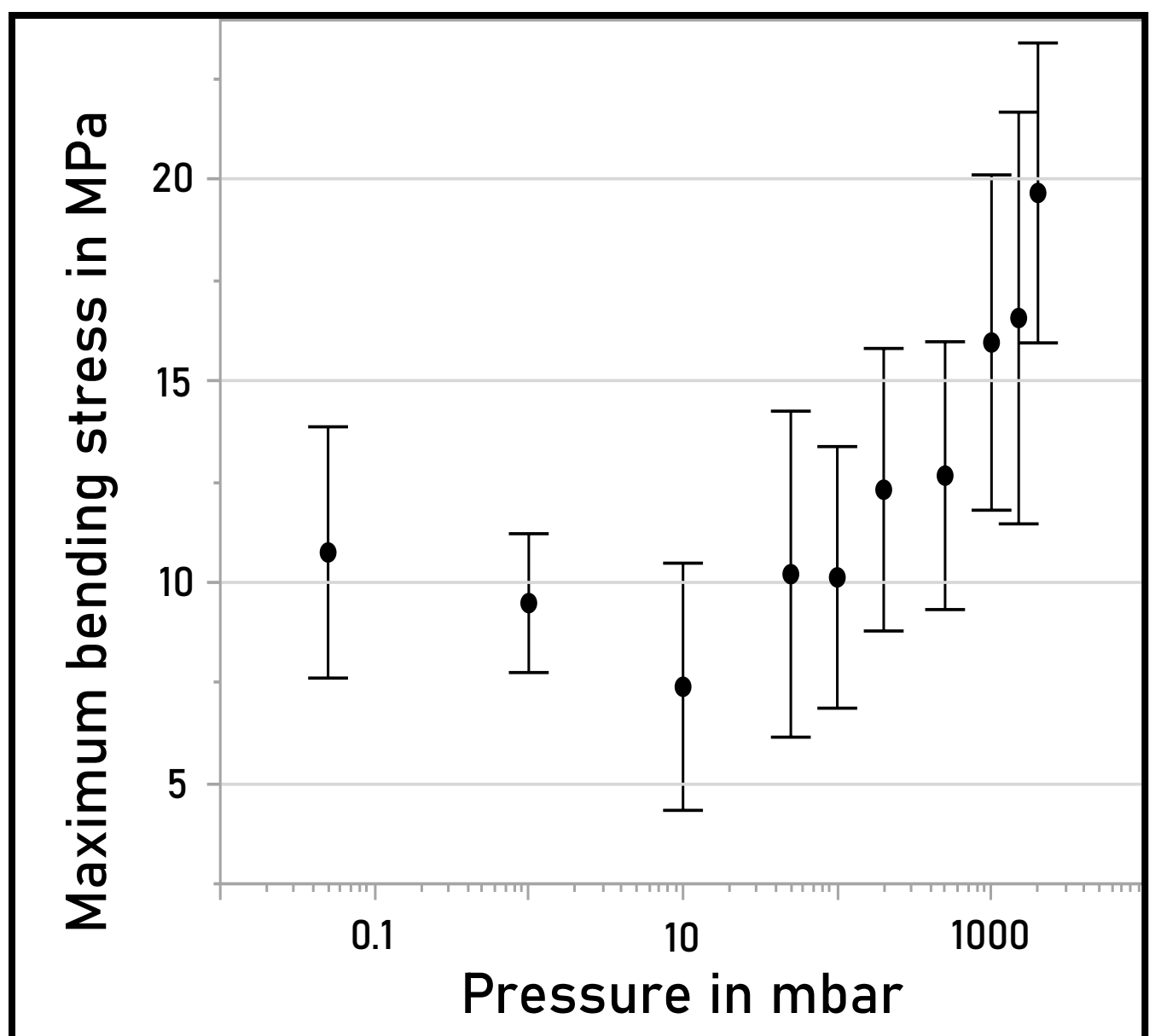
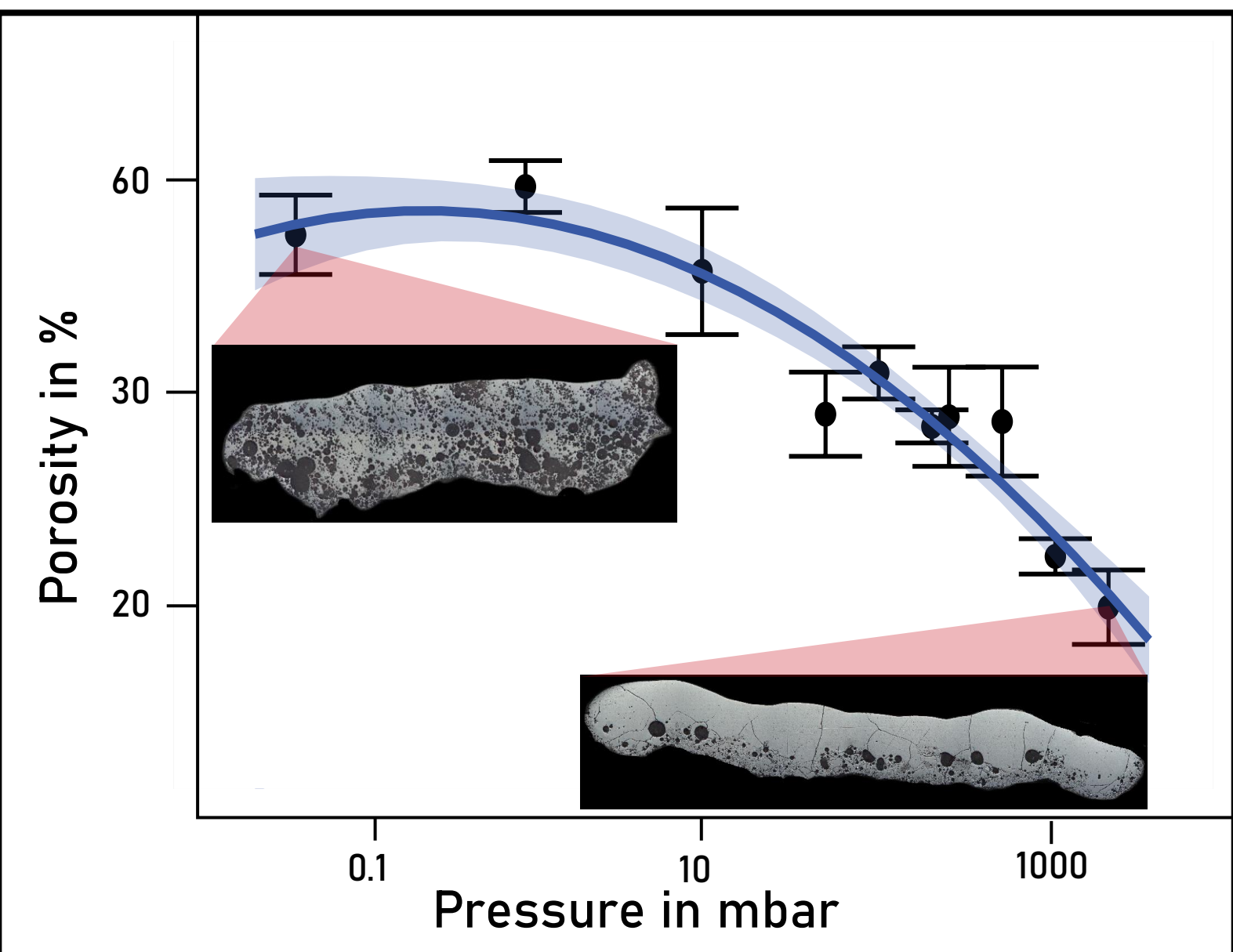
▲ Laser melting setup for sample fabrication on gondola of Einstein-Elevator



▲ Einstein-Elevator principal components



◀ Turntable for sample fabrication holding 20 regolith powder beds, used in Einstein-Elevator to produce samples at lunar gravity



▲ Porosity (left) and maximum bending stress according to DIN EN 843 standard (right) of samples fabricated from LX-I50 regolith simulant as a function of ambient pressure [4]

References

- C. Lotz, et al. Einstein-Elevator: A New Facility for Research from μg to $5g$. Gravitational & Space Research 5 (2) (2017).
- L. Overmeyer, et al. Laser powder directed energy deposition and substrate-free single layer powder bed fusion under micro- and lunar gravity conditions. CIRP Annals – Manufacturing Technology (2025).
- J. Patzwald, et al. Properties of novel LX lunar regolith simulant system – The base simulants: Part 1. Acta Astronautica 231 (2025).
- T. Griemsmann, et al. Influence of ambient pressure on laser beam melting of lunar regolith simulant. Acta Astronautica 228 (2025).